Image-Based Measurement Laboratory
Lab1 - Camera Basics

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Teaching Goals

This first laboratory serves to introduce you to the basics of image acquisition. You will explore the effects of shutter speed, aperture, sensor gain as well as focal length. Additionally you will get to know a way to undistort images.

Important Note!

Please strictly adhere to the following rules when working in the laboratory:

- **Never** use a camera without a suitable mount such as a tripod!
- **Never** touch the unprotected surface of a sensor!

Exercise 1 - Aperture, Shutter Speed, Gain

Aperture, shutter speed and sensor gain are all parameters than can be varied in order to obtain a proper exposure. Generally speaking a proper exposure means that the full value range of the image (usually 0-255, `uint8`) is used without saturating at either end of the range. How does each of these parameters relate to the exposure?

Also, each of the aforementioned parameters comes with its own unique effects. This means that while you can vary the exposure equally with each parameter, the visual quality of the resulting images may not be the same. To explore those effects, record properly exposed and sharp images of a (dynamic?) scene using low and high values of each parameter.

Describe your findings and explain why the observed effects occur.

Exercise 2 - Auto-Focus

Take a sequence of about $N = 10$ images of a static scene with varying focus. Determine scene properties which make the scene well suited for an auto-focus algorithm and also think about how to best record the images (remember Exercise 1).

Develop a MATLAB script to determine the most in-focus image out of this sequence. Provide a plot that quantifies how much each image is in focus.

Exercise 3 - Dolly-Zoom

Is zooming (i.e. increasing the focal length) the same as moving closer to the subject?

Try to come up with a formula to calculate the field of view for a given camera and focal length. Then record at least two images of a scene with different focal lengths (eg. 6mm and 12mm) such that an object in the scene (eg. a checkerboard target) is framed equally in both images - *Note: The target should occupy no more than a quarter of the area of the frame.* What do you observe?

Exercise 4 - Removing projective distortion

Record an image of an A4 sheet (that you could and should write or draw upon beforehand) at an oblique angle. Then undistort the image by calculating a homography, such that a straightened version of the A4 sheet is obtained. Try this for different angles.

Bonus: Also record an image of a building with your phone and undistort the front of the building.